Accelerated Decline in Hepatic Glucose Production During Fasting in Normal Women Compared With Men

John N. Clore, Penny S. Glickman, Shirley T. Helm, John E. Nestler, and William G. Blackard

Plasma glucose values have been reported to be lower in women than in men after a 72-hour fast. However, a comparison of glucose kinetics in fasting men and women has not been described. Therefore, five normal men and five normal women underwent sequential 3-^{3}H-glucose infusions after both a 14- and a 64-hour fast. Plasma glucose levels fell similarly during the fast in men (5.23 ± 0.03 *v* 3.96 ± 0.14 mmol/L, $P < .01$) and women (4.84 ± 0.14 *v* 3.65 ± 0.25 mmol/L, $P < .01$). The fall in plasma glucose was associated with a significantly greater fall in glucose appearance (Ra) in women compared with men ($P < .05$). Ra fell 15.8% ± 3.0% in men (2.11 ± 0.24 to 1.79 ± 0.24 mg.kg^{-1}.min^{-1}, $P < .01$) and 24.6% ± 1.4% in women (2.22 ± 0.17 to 1.67 ± 0.12 mg.kg^{-1}.min^{-1}, $P < .001$). During the fast, plasma glycerol, free fatty acids (FFA), and β-hydroxybutyrate levels rose significantly and plasma alanine fell significantly in both sexes. Plasma glycerol levels were significantly higher in women compared with men after fasting (0.16 ± 0.01 *v* 0.11 ± 0.02 mmol/L, $P < .05$). In addition, the transition from ambulation to bed rest demonstrated unexpected sex-related differences in glucose homeostasis after the 64-hour fast. During the two-hour equilibration period required for glucose kinetic studies (subjects reclining), significant decrements in glucose, FFA, and lactate were observed in the 64-hour fasted women but not in the men. In conclusion, women exhibit a significantly greater fall in Ra after 64 hours of fasting, which may account for the reportedly greater hypoglycemia during fasting in women. The failure to demonstrate a significantly lower plasma glucose level in women after a 64-hour fast in the present study may be related to the ambulatory status of our subjects.
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METABOLIC STUDIES often include data obtained from both men and women without comment on potential sex-related differences. Following an overnight fast (14 hour) plasma glucose levels in men and women are similar.[1-4] However, the nadir plasma glucose value is lower in women compared with men following a short-term (72 hour) fast, when subjects are studied at rest.[1-4] A satisfactory explanation for the sex-related difference in glucose homeostasis is not available. The greater fall in plasma glucose has been described in both the midfollicular and midluteal phase of the menstrual cycle and is not reproduced by the administration of estrogens to normal men.[2] The lower plasma glucose concentration observed in women has been associated with either unchanged[1] or higher[2,3] plasma glucagon levels, lower glucogenic amino acid levels,[1] and increased rates of lipolysis and ketone body formation[1-3] when compared with men. These latter changes may reflect the reduced muscle mass and greater amounts of adipose tissue in women.[5]

Clearly, the lower fasting plasma glucose in women compared with men must be due to either a decrease in hepatic glucose output (HGO) and/or an increase in glucose utilization. Studies in obese[6] and normal[7,8] men following a prolonged fast have shown a fall in HGO as glycogen stores are depleted and the contribution of gluconeogenesis to total glucose output increases.[8] The present study was designed to examine glucose kinetics (glucose utilization and hepatic glucose output) after an overnight and a three-day fast in normal men and women. Changes in hormonal and metabolic parameters that may affect glucose homeostasis were also determined.

MATERIALS AND METHODS

Ten normal volunteers were admitted to the General Clinical Research Center (GCRC) at the Medical College of Virginia for the metabolic studies. Five of the subjects were male and five were female in the midfollicular phase of their menstrual cycle. None had a history of diabetes mellitus, hypertension, or kidney disease or were on any medications. Subjects were matched for age and body mass index (BMI). Body composition was assessed by skinfold measurements and impedance plethysmography (BIA-101 Body Composition Analyzer, RJL Systems, Detroit), which closely reflects more conventional methods.[9,10] Determination of body fat by the two methods was closely correlated ($r = .94$, $P = .005$), and the values reported are the mean of the two methods (Table 1).

Following a meal at 6 PM the subjects were fasted until completion of the study (total fast 64 hour). Subjects were at bed rest during the three hours of each study period but were permitted to leave the GCRC to carry out their usual daily activities in the interim. The activity level of the men and women was not appreciably different and none of the subjects participated in vigorous exercise during the study period. Baseline blood samples were obtained at 7 AM (after 13 hours of fasting) and the subjects were then put to bed rest. A primed (25 μCi) continuous (0.25 μCi) infusion of 3-^{3}H-glucose (New England Nuclear, Boston) was administered via a forearm vein from 7 AM until 10 AM for the determination of glucose specific activity. An indwelling venous catheter was inserted in the contralateral forearm and was kept patent with an infusion of 0.9% NaCl. No heparin was used in the studies and no blood samples were obtained during the two-hour equilibration period from 7 AM to 9 AM, during which time steady state was achieved. Thereafter, blood samples were obtained at 20-minute intervals during the last hour of the study (9 AM to 10 AM) for the determination of glucose specific activity and measurement of plasma hormones, lactate, alanine, β-hydroxybutyrate, glycerol, and free fatty acids (FFA). After completion of the first study (10 AM), the fast was continued for an additional 48 hours, at which time the above procedure was

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Table 1. Demographic Data of Normal Male (n = 5) and Female (n = 5) Volunteers During a 64-Hour Fast

	Men	Women
Age (yr)	25 ± 1	29 ± 4
Wt (kg)	79.9 ± 8.3	62.8 ± 5.0
BMI (kg/m^2)	25.1 ± 2.5	23.7 ± 1.0
% Fat	16.6 ± 2.6	29.2 ± 1.0*
Weight loss (kg)	3.18 ± 0.6	2.26 ± 0.25

*$P < .01$ *v* men.

repeated. Compliance with the fast was assessed by urinary ketone determinations and history.

Plasma glucose was measured immediately after centrifugation by the glucose oxidase method (Beckman, Fullerton, CA). Plasma insulin,[11] C-peptide,[12] growth hormone,[13] and glucagon[14] were determined using double-antibody radioimmunoassays. Cortisol was measured by direct radioimmunoassay (Gammacoat assay, Clinical Assays, Cambridge, MA). Plasma FFA was determined by enzymatic methods.[15] Blood samples for determination of glucose specific activity and metabolites were immediately deproteinized with ice-cold 3 mmol/L perchloric acid. The supernatant was neutralized with 3 mmol/L KOH and the resulting supernatant assayed for l-lactate,[16] glycerol,[17] alanine,[18] and β-hydroxybutyrate.[18] Because plasma glucose was constant during the time period (third hour of 3-^{3}H-glucose infusion) during which glucose kinetics were being measured, rates of glucose appearance (Ra) and disappearance (Rd) were calculated by dividing the rate of glucose infusion by the steady state glucose specific activity. (The coefficients of variation for glucose specific activity during the 16th and 64th hour of each study were 0.025, 0.028, 0.041, and 0.032 for the women and men, respectively). Glucose kinetics calculated by Steele's non–steady-state equation gave identical results. Because no glucose was infused during the studies, Ra may be equated with HGO. Glucose clearance (MCR) was calculated by dividing the rate of glucose disappearance (Rd) by the plasma glucose value.

For each subject the four values obtained during the final hour of each study (the 16th and 64th hours of the fast) were averaged and used for statistical analysis. All remaining values represent measurements at a single time point. Data were managed and analyzed using the CLINFO data analysis system (BBN Software Products, Cambridge, MA). Results are expressed as the mean ± SEM. Data were tested for normality by the Wilk-Shapiro test. Comparisons between the two groups (women *v* men) were performed using Student's two-tailed unpaired *t* test, and within groups using Student's two-tailed paired *t* test. Statistical significance was assumed when $P < .05$.

RESULTS

During the 64-hour fast, mean weight loss was similar in men and women [3.18 ± 0.6 (3.9%) and 2.26 ± 0.25 kg, (3.6%) NS, Table 1]. Plasma glucose values fell by 24% from the 16th to the 64th hour of fasting in both men (5.23 ± 0.03 *v* 3.96 ± 0.14 mmol/L, $P < .001$) and women (4.84 ± 0.14 *v* 3.65 ± 0.25 mmol/L, $P < .001$). Although the plasma glucose was consistently lower in women compared with men, these differences were not statistically significant. Rates of glucose appearance (Ra) fell from 2.11 ± 0.24 to 1.79 ± 0.24 mg.kg^{-1}.min^{-1} ($P < .01$), and 2.22 ± 0.17 to 1.67 ± 0.12 mg.kg^{-1}.min^{-1} ($P < .01$) in men and women, respectively (Fig 1). Both the absolute (0.55 ± 0.06 *v* 0.32 ± 0.06 mg.kg^{-1}.min^{-1}) and the percent fall in Ra (24.6% ± 1.4% *v* 15.8% ± 3.0%, Fig 1, inset) were significantly greater in women compared with men ($P < .05$). Despite the greater fall in Ra in women, their plasma glucose values were not significantly lower than in men after the 64-hour fast due to a corresponding fall in Rd. Glucose clearance rates did not change significantly in either sex after the 64-hour fast (Fig 1).

Changes in plasma glucose were associated with decreases in plasma insulin and C-peptide as well as increases in plasma glucagon (Table 2). Mean growth hormone levels increased insignificantly in the men and women (Table 2), and plasma cortisol levels were unchanged after the 64-hour fast. Plasma catecholamine levels were not determined.

Measurements of plasma intermediary metabolites at the end of 14 and 64 hours of fasting are shown in Table 3. The expected rise in glycerol concentrations in men (0.05 ± 0.01 *v* 0.11 ± 0.02 mmol/L, $P < .05$) and women (0.07 ± 0.01 *v* 0.16 ± 0.01 mmol/L, $P < .01$) was observed. Moreover, plasma glycerol levels were significantly greater in women compared with men after the 64-hour fast ($P < .05$). Plasma FFA also rose significantly in men (0.49 ± 0.06 *v* 0.81 ± 0.09 mmol/L, $P < .01$) and women (0.47 ± 0.04 *v* 0.89 ± 0.04 mmol/L, $P < .05$), without significant intersex differences. Plasma alanine concentrations fell similarly in the men and women (Table 3), but plasma lactate did not change significantly after the 64-hour fast in either sex. Because the subjects were allowed off the GCRC, the possibility exists that compliance with fasting may have been less than complete. Although plasma β-hyroxybutyrate levels varied considerably both before and after the fast (Table 4), all subjects exhibited at least a sixfold increase during the fast. Furthermore, no correlation was found between peak

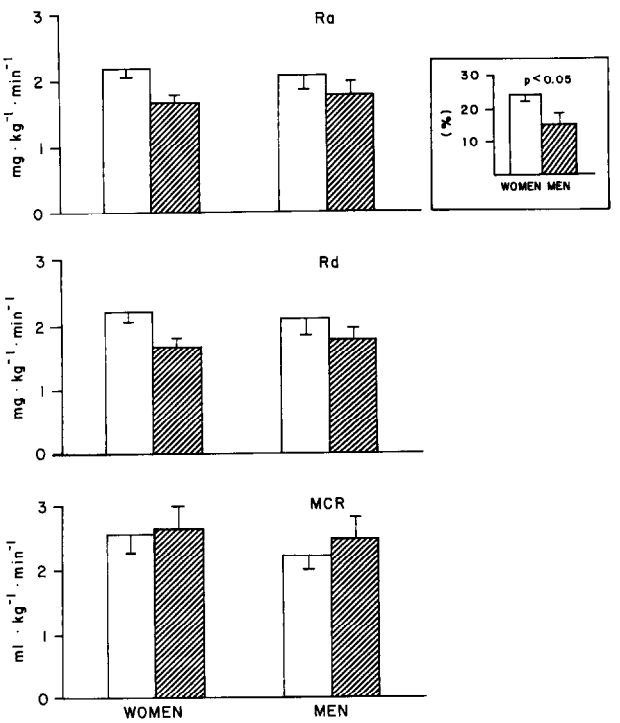

Fig 1. Comparison of glucose production (Ra), glucose utilization (Rd), and metabolic clearance rate (MCR) in normal women (n = 5) and men (n = 5) during the 16th (□) and 64th (▨) hours of fasting. (Inset), Percent decrease in Ra from 16th to 64th hour of fasting in normal women and men.

Table 2. Plasma Glucose and Glucoregulatory Hormone Concentrations During the 16th and 64th Hour of a Fast in Normal Men and Women

	Men		Women	
	16 Hour	64 Hour	16 Hour	64 Hour
Glucose (mmol/L)	5.23 ± 0.03	3.96 ± 0.14*	4.84 ± 0.14	3.65 ± 0.25*
Insulin (mU/L)	6.1 ± 0.8	4.6 ± 0.2	8.0 ± 2.0	5.2 ± 0.6
C-Peptide (nmol/L)	0.46 ± 0.07	0.29 ± 0.03†	0.53 ± 0.07	0.33 ± 0.03†
Glucagon (pmol/L)	40.9 ± 4.2	68.5 ± 5.0†	37.9 ± 4.8	52.9 ± 7.4†
Cortisol (μmol/L)	0.33 ± 0.04	0.35 ± 0.04	0.26 ± 0.04	0.26 ± 0.04
Growth hormone (nmol/L)	0.09 ± 0.04	0.19 ± 0.09	0.06 ± 0.04	0.13 ± 0.07

*$P < .01$ *v* 16-hour value.
†$P < .05$ *v* 16-hour value.

β-hydroxybutyrate levels and the fall in Ra ($r = .21$, $P = .56$). However, significant changes in glucose metabolism were observed during the two-hour transition from ambulation to bedrest. Following the 64-hour fast, values for plasma glucose, FFA, and lactate consistently fell during the 120-minute equilibration period. As shown in Fig 2, plasma glucose fell from 4.25 ± 0.23 to 3.65 ± 0.25 mmol/L ($P = .05$) after the ambulatory women were recumbant for 120 minutes. The decrease in plasma glucose was associated with significant falls in FFA (1.30 ± 0.12 to 0.89 ± 0.04 mmol/L, $P = .02$) and lactate (2.05 ± 0.34 to 1.10 ± 0.07 mmol/L, $P = .03$). This effect was more pronounced, achieving statistical significance, in women compared with men, and was not observed in either sex after the overnight fast (not shown).

DISCUSSION

This investigation was designed to examine the mechanisms responsible for the greater fall in plasma glucose in women compared with men after a short-term fast. During the initial phase of fasting, after glycogen stores are depleted, and before oxidation of carbohydrates is replaced by oxidation of FFA, enhanced hepatic gluconeogenesis is required to prevent the development of severe hypoglycemia. After a three-day fast, the contribution of gluconeogenesis to overall glucose output increases from 25% to 95%.[8] Because of the role of peripheral metabolites as either substrates (amino acids, lactate, and glycerol) for or promoters (FFA) of gluconeogenesis, known differences in body composition between men and women[5] might influence glucose homeostasis during fasting. We postulated that these differences might be responsible for the lower plasma glucose values reported in women after a three-day fast. Furthermore, previous studies examining this issue have involved subjects who were maintained at bed rest during the period of blood sampling.[2,3] Therefore, we also examined the possibility that activity level (eg, ambulation *v* bed rest) might influence glucose homeostasis during fasting.

As expected, plasma glucose, Ra, and Rd fell significantly in both men and women after a three-day fast compared with the overnight fast. However, our studies did not detect significantly lower plasma glucose values in women compared with men after either an overnight (16 hour) or more prolonged (64 hour) fast. The inability to demonstrate a statistically significant difference between the two sexes, as has been observed by others,[1-4] may be due in part to the limited number of subjects studied. However, it may also be due to the apparent influence of activity level on glucose homeostasis in women compared with men (Fig 2). After 61 hours of fasting, subjects arrived on the GCRC ambulatory and were put to bed rest. During this period plasma glucose, FFA, and lactate declined significantly in the women, whereas no change was noted in the men. Other investigators have also observed a greater effect of activity level on plasma FFA levels in women compared with men.[19]

It is likely that the higher levels of FFA and lactate in women in the ambulatory state promoted and sustained a higher rate of HGO than in the nonambulatory state. Plasma concentrations of FFA and lactate declined significantly after the women were put to bed rest, and presumably HGO declined as well. It is possible that, had plasma glucose levels been measured after subjects had been kept at bed rest for an even longer period, plasma glucose levels would have been significantly lower in women than in men. The fact that plasma glucose values were similar in men before and after the equilibration period suggests that the ambulatory state probably perturbed their glucose kinetics less.

Table 3. Plasma Concentrations (mmol/L) of Glycerol, Free Fatty Acids (FFA), β-Hydroxybutyrate (β-OH-B), Lactate, and Alanine During the 16th and 64th Hours of a Fast in Normal Men and Women*

	Men		Women	
	16 Hour	64 Hour	16 Hour	64 Hour
Glycerol	0.05 ± 0.01	0.11 ± 0.02†	0.07 ± 0.01	0.16 ± 0.01‡
FFA	0.49 ± 0.06	0.81 ± 0.09‡	0.47 ± 0.04	0.89 ± 0.04‡
β-OH-B	0.16 ± 0.06	2.54 ± 0.27‡	0.14 ± 0.03	3.16 ± 0.80‡
Lactate	1.09 ± 0.09	1.23 ± 0.10	1.27 ± 0.03	1.11 ± 0.07
Alanine	0.40 ± 0.03	0.26 ± 0.01†	0.41 ± 0.04	0.32 ± 0.02†

*Plasma samples were obtained after two hours at bed rest (9 AM to 10 AM).
†$P < .05$ *v* 16-hour value.
‡$P < .01$ *v* 16-hour value.

Table 4. Comparison of Glucose Appearance (Ra) and β-Hydroxybutyrate (β-OHB) in the 16th and the 64th Hour of a Fast in Normal Men (Subjects 1 to 5) and Women (Subjects 6 to 10)

Subject No.	Ra (mg · kg^{-1} · min^{-1}) 16 Hour	Ra (mg · kg^{-1} · min^{-1}) 64 Hour	βOHB (mmol/L) 16 Hour	βOHB (mmol/L) 64 Hour
Men				
1	2.66	2.40	0.42	2.91
2	2.04	1.70	0.08	1.54
3	1.33	1.15	0.08	3.09
4	2.58	2.27	0.12	2.51
5	1.96	1.43	0.08	2.65
Mean ± SEM	2.11 ± 0.24	1.79 ± 0.23*	0.16 ± 0.06	2.54 ± 0.27*
Women				
6	1.97	1.41	0.17	1.44
7	1.73	1.39	0.08	1.51
8	2.31	1.77	0.09	3.41
9	2.36	1.73	0.26	5.96
10	2.75	2.07	0.09	3.47
Mean ± SEM	2.22 ± 0.17	1.67 ± 0.12*	0.14 ± 0.03	3.16 ± 0.80*

*$P < .01$ *v* 16-hour value.

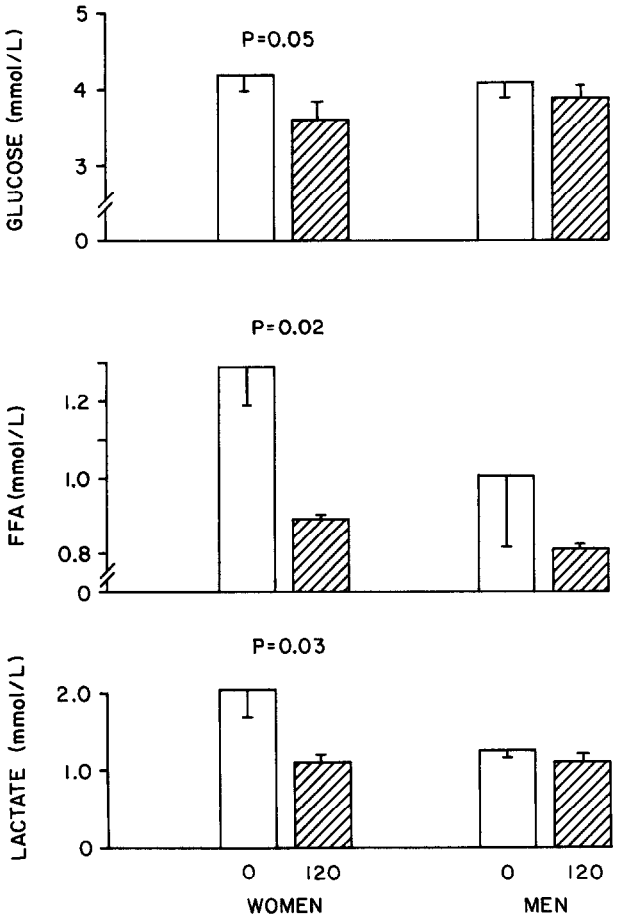

Fig 2. Effect of 120 minutes of bed rest on plasma glucose, free fatty acids (FFA), and lactate in normal women and men after a 64-hour fast. Samples were obtained on admission in the ambulatory state (0 min, □) and after a 120 min (▨) equilibration period while the subjects were at bed rest. Results are expressed as mean ± SEM.

Although the present study demonstrates a significantly greater fall in Ra during a 64-hour fast in women compared with men, a ready explanation for this finding is not available. No sex-related differences in glucoregulatory hormone concentrations were observed in our study (Table 2). Measured substrates for gluconeogenesis (lactate, alanine, and glycerol) were either the same or higher in women and levels of gluconeogenic promoters (FFA and glucagon) were similar in both sexes. The finding of increased plasma glycerol levels in women compared with men after 64 hours of fasting supports the increased rates of lipolysis in women suggested by Merrimee and Fineberg,[2] and probably reflects their higher body fat composition.[5] Levels of total glucogenic amino acids (not measured in the present study) are reportedly lower in women compared with men after a short-term fast[1] and may lead to lower rates of glucose production when other gluconeogenic substrates are limited. During ambulation the provision of lactate (and glycerol) derived from both muscle glycogen and adipose tissue[20,21] may have provided sufficient substrate for new glucose production to prevent hypoglycemia in the women. However, when the production of these substrates diminishes after the two-hour reclining period, a fall in Ra would be anticipated and gluconeogenesis might then depend more on total glucogenic amino acids.

In conclusion, 64-hour fasted women exhibit a greater fall in HGO than do men similarly fasted. The failure to demonstrate significantly depressed plasma glucose values in women in this study may have been related to the enhanced availability of gluconeogenic substrates (eg, lactate) and promoters (FFA) during ambulation. When put to bed rest, plasma levels of these metabolites, as well as of glucose, fell significantly in women. In contrast, no change was observed in men.

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